A shared action coefficient from classical composition

A nonnegative mass-dependent fluctuation coefficient becomes mass independent when independent composition preserves its value as a whole-body observable. A single finite-rate, nonzero-speed reference constituent then makes the shared coefficient strictly positive. This conditional result separates the composition premise from the physical clock that sets the value.

1. Observable and independent composition

All masses are positive, positions are on the line, and coefficients have action units. For centered independent displacement variables over a common window $\Delta > 0$, define $\mathrm{Var}(\Delta X_i) = \kappa_i \Delta / m_i$. Let $M = m_1 + m_2$, $\mu = m_1 m_2 / M$, $R = (m_1 X_1 + m_2 X_2)/M$ and $r = X_1 - X_2$. Variance addition gives

$$\kappa_{\rm cm} := \frac{M}{\Delta}\,\mathrm{Var}(\Delta R) = \frac{m_1\kappa_1 + m_2\kappa_2}{M}, \qquad \kappa_{\rm rel} := \frac{\mu}{\Delta}\,\mathrm{Var}(\Delta r) = \frac{m_2\kappa_1 + m_1\kappa_2}{M}.$$

Thus $\kappa_{\rm cm} + \kappa_{\rm rel} = \kappa_1 + \kappa_2$ and $\mathrm{Cov}(\Delta R, \Delta r) = \Delta(\kappa_1 - \kappa_2)/M$. For independent Gaussian bridges replace Δ by their common covariance kernel; the same coefficient transformation holds. In that Gaussian setting, centre/relative process independence is equivalent to $\kappa_1 = \kappa_2$.

For independent stationary finite irreducible reversible chains J^i, take centered velocities $v_i(J^i)$ and $X_i(t) = \int_0^t v_i(J^i_s)ds$. C019 defines $H_i = 2m_i \int_0^\infty C_i(s)ds$. Independence gives $C_{\rm cm} = (m_1^2 C_1 + m_2^2 C_2)/M^2$, so the same formulas hold with H in place of κ in the long-window limit. The product state has generator $Q_1 \otimes I + I \otimes Q_2$ and invariant law $\pi_1 \otimes \pi_2$; detailed balance and irreducibility hold factorwise. Retain this state even if distinct state pairs share one velocity value: the projected velocity alone need not be Markov. If each constituent speed is at most $u < c$, the centre speed has the same bound. Relative velocity may reach $2u$; it is a coordinate difference, not a separate particle speed premise.

2. Mass-universality theorem

Assumptions. Fix a preparation class. Every mass $m \in (0, \infty)$ has a finite coefficient $\kappa(m) \geq 0$ depending only on its mass within this class. Independent constituents can be composed, and the coefficient assigned to their whole, of mass M, equals the centre coefficient calculated above. Choose either the common-window coefficient, a Gaussian coefficient, or the Green–Kubo plateau throughout; the theorem does not exchange these observables.

Conclusion. There is a constant $K \geq 0$ such that $\kappa(m) = K$ for every $m > 0$.

Proof. Set $f(m) = m\kappa(m)$. Composition says $f(a+b) = f(a) + f(b)$ on positive reals. For $b > a$, nonnegativity gives $f(b) - f(a) = f(b-a) \geq 0$. Fix a mass unit $m_0 > 0$. Additivity yields $f(qm_0) = qf(m_0)$ for every positive rational q. For any $m > 0$, choose rationals $q_n \uparrow m/m_0$ and $r_n \downarrow m/m_0$. Monotonicity sandwiches $f(m)$ between $q_n f(m_0)$ and $r_n f(m_0)$. Hence $f(m) = mf(m_0)/m_0$, and $K = f(m_0)/m_0$. This also covers $K = 0$.

The premise equating independently composed and whole-body preparations does the physical work. All-positive-mass admissibility makes the rational argument available; a restricted species list needs a separately stated domain theorem.

3. Positive reference theorem

Apply the preceding theorem to Green–Kubo plateaus. Suppose the class contains one reference constituent of mass m_0, stationary symmetric velocities $\pm u_0$, with $0 < u_0 < c$ and reversal rate $0 < \lambda_0 < \infty$. C020 gives

$$\boxed{K = H(m_0) = \frac{m_0 u_0^2}{\lambda_0} > 0, \qquad H(m) = K \quad (m > 0).}$$

This excludes an attained zero in that preparation class using classical probability, nonzero motion, a finite positive clock and the composition premise. The value comes from the reference constituent. Other two-state members consequently obey $\lambda_m = mu_m^2/K$.

For any chosen $K > 0$ and common $0 < u < c$, take independent two-state members with $\lambda_m = mu^2/K$ and include all their finite products. Every product has the same plateau when normalized by its total mass. This realizes coefficient closure even though its centre velocity generally has more than two values. It is a consistency example for the assumptions, not a derivation of the assigned rate law. Across such examples K can approach zero while each individual rate remains finite. A class-wide numerical lower bound needs uniform controls, for example $u_0 \geq u_{\min} > 0$ and $\lambda_0 \leq \Lambda < \infty$ at fixed m_0.

4. Preparation and correlation tests

An environmental label θ allows $\kappa(m, \theta) = K(\theta)$ under the same composition proof at each fixed θ. Universality across such classes requires a preparation-equivalence premise. If cross covariance is $C_{12} = \mathrm{Cov}(\Delta X_1, \Delta X_2)$, the centre coefficient has an additional $2m_1m_2C_{12}/(M\Delta)$; independence fixes it to zero. Bound bodies and common reservoirs must be tested with their actual correlations.

In A02, separately prepared tracers in otherwise identical refreshed baths have plateau $H(m) = (m + M_b)s^2/\nu$. Treating their total mass as one tracer changes the coefficient by

$$H(m_a + m_b) - \frac{m_a H(m_a) + m_b H(m_b)}{m_a + m_b} = \frac{2m_a m_b s^2}{\nu(m_a + m_b)}.$$

The positive discrepancy for $s > 0$ identifies failure of the preparation equivalence premise, while preserving the earlier bath calculation. The comparison is between these specified experiments; dynamics of an actual bound composite require their own mechanical model.

5. Scope, source chain and next test

The theorem establishes a conditional universal positive **long-window coefficient**, with action units and a specified preparation class. Its role as a quantum phase constant and its physical-time attraction are subsequent obligations. At bounded speed the microscopic variance coefficient still tends to zero by C018. The next A09 calculation connects these separate windows to the real telegraph and complex checkerboard propagators.

This note develops the user's composition suggestion and the independent P02 product-chain calculation. C019–C020 supply the reviewed correlation input; Pavliotis (2010) gives its Poisson-equation source, and Pitman–Yor (2018) the Gaussian covariance background. Per-result prior-art status belongs to B16. The five dedicated algebra checks cover the two coefficients, their sum, their cross covariance and the bath discrepancy. Existing P02 checks also test a nine-state product generator. The written proofs supply the general additive-function and positive-reference conclusions.